



An efficient stacking-based ensemble technique for early heart attack prediction

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Abstract

In many parts of the world, heart disease is the leading cause of death. Preventing or effectively managing cardiac disease often depends on its early detection. There has been a significant uptick in research towards using machine learning to estimate the probability of cardiovascular disease. Using a variety of classification methods and stacking as ensemble techniques, this work investigates the problem of predicting cardiovascular illness. A total of 1025 patients are used in the analysis, and their clinical data is broken down into 14 different categories (e.g., age, sex, chest pain kind, blood pressure, cholesterol levels, and more). The initial step of the analysis is to preprocess the data by filling in missing values, standardizing the numbers, and encoding the categories. After that, the information is segmented into a training set and a test set for the purposes of model building and testing. Logistic Regression, Decision Tree, Random Forest, Extreme Gradient Boost, Naive Bayes, and K-Nearest Neighbors (KNN) are the six classification methods used in the research. Accuracy, precision, recall, and F1-score are only some of the measures used to assess the efficacy of various classification methods. The findings reveal that random forest and decision tree both yields a 92.68% accuracy, with extreme gradient boost coming in as a close second at 90.73%. In the second portion of the research, ensemble approaches, and more especially stacking, are used to boost the classification models' accuracy. The goal of stacking, a method of ensemble machine learning, is to increase prediction precision by using numerous models in concert. By training a meta-classifier on the predictions of the base models, stacking combines the predictions of multiple base models. The results demonstrate that stacking considerably enhances the efficiency of the original classifiers. The stacked model outperforms each individual classifier, with an accuracy of 98.53%.

Keywords Stacking · Ensemble learning · Heart attack · XGBoost · Meta-classifier · Features

1 Introduction

Heart disease is a group of conditions that affect the heart. The World Health Organization says that 17.9 million people die every year from heart disease, making it the top cause of death in the world. Heart disease risk goes up with bad habits like having high cholesterol, being overweight, having high blood pressure, and having high triglyceride levels. The American Heart Association lists a number of symptoms, such as trouble sleeping, irregular heartbeats, swollen legs, and fast weight gain, that can also be signs of other diseases in older people. This makes it hard to make a correct diagnosis. Thanks to the large amount of study data and hospital patient records that are available, doctors can make accurate diagnoses and find heart disease before it kills the person [1].

Heart disease is hard to spot because there are many things that can put you at risk, like diabetes, high blood pressure, an odd pulse rate, and others. Data mining and neural networks have been used with different classification methods like K-Nearest Neighbour Algorithm, Decision Trees, Genetic algorithm, and Naive Bayes to figure out how bad heart disease is. Because heart disease is so complicated, it needs to be treated carefully to avoid bad health effects, like dying too soon [2].

With more medical data being collected, doctors can use computer technology to better diagnose patients. In the healthcare business, machine learning is becoming an increasingly important way to help people make decisions. This analytical tool can help with big, complicated jobs that are hard to programme, like turning medical records into knowledge, predicting pandemics, and looking at genomic data. The high number of datasets is a big problem for machine learning because it takes a lot of memory to look at a lot of features and can lead to overfitting. To solve this problem, feature weighting is used to cut down on unnecessary data and processing time, which improves the algorithm's performance in the long run. In many areas, like health management, genome expression analysis, medical images, and the Internet of Things (IoT) [3], it is important to find a small set of features that can be used to describe different illnesses. Dimensionality reduction methods, such as feature extraction and feature selection, are used to make data easier to understand and change by getting rid of useless features [4].

In the past, doctors and traditional statistical methods were used to figure out how likely a sickness was to happen and what the outcome would be. But this method often led to bias, mistakes, high prices, and lower-quality care for patients. With more electronic health data becoming available, more advanced computer techniques, like machine learning, have become more useful and are used more often in the area of disease prediction [5].

In order for machine learning algorithms [6] to work well, they must be taught on the right sets of data. Techniques for selecting features like data mining, Relief selection, and LASSO can be used to prepare the data for more accurate forecasts by making sure the training and test data are the same. Once the important traits have been chosen, classifiers and hybrid models can be used to figure out how likely it is that a disease will happen. But problems like restricted medical datasets, feature selection, the use of machine learning (ML) algorithms, and not enough analysis may make it hard to predict heart disease accurately. There are different ways to deal with these problems [9]. In the first step of this study, the data is preprocessed to handle missing values, code categorical variables, and scale it correctly. After that, the data is split into training and testing sets, which are used to build and test the models. In this study, six different ways of classifying things are used: Logistic Regression, Decision Tree, Random

Forest, Extreme Gradient Boost, Naive Bayes, and K-Nearest Neighbours (KNN). In the second step, researchers use ensemble methods, especially stacking, to improve the performance of classification models.

Different metrics, like accuracy, precision, recall, and F1-score, are used to measure how well each classification method works. The results show that the most accurate method is extreme gradient boost, with 93.17%, followed by Decision Tree, with 92.68%. In the second part of the study, ensemble methods, especially stacking, are used to improve how well the classification models work. Stacking is a type of machine learning called “ensemble learning.” It combines various models to make a prediction more accurate. In stacking, the predictions of several base models are joined using a meta-classifier that is trained on the predictions of the base models. The results show that stacking makes the base algorithms much better at what they do. The stacked model has a higher accuracy than any of the individual predictors, at 96.58%.

1.1 Motivation

The prevalence of heart disease is rising, especially in emerging nations, and it is a leading cause of death worldwide. Heart disease is a broad term that encompasses a variety of cardiac-related illnesses, such as coronary artery disease, heart failure, arrhythmias, and heart valve disorders. Heart attacks, strokes, and even death are among the possible outcomes of these disorders. Heart disease is a leading cause of death and disability, although it can be mitigated via early diagnosis and preventative measures. Patients’ health and survival rates can be improved via early detection and prevention of cardiovascular disease. However, conventional risk assessment techniques for cardiovascular disease prediction, including the Framingham risk score, have shortcomings in properly forecasting individual risk. Machine learning and artificial intelligence approaches offer a promising way to improving cardiac disease prediction in light of the availability of enormous volumes of health data, such as electronic health records and wearable devices. Researchers can use this information to create models that properly predict who will develop heart disease and who should receive individualized preventative care.

1.2 Research objective

A heart disease prediction research project has the potential to have a significant impact on public health by improving the accuracy of heart disease prediction and management. By identifying individuals at high risk of developing heart disease, healthcare providers can intervene earlier, potentially preventing adverse health outcomes. Additionally, by personalizing preventive interventions based on an individual’s risk profile, healthcare providers can optimize treatment and improve patient outcomes. Furthermore, a heart disease prediction research project can help reduce the economic burden of heart disease on healthcare systems worldwide. By improving the accuracy of heart disease prediction, healthcare providers can optimize the use of healthcare resources, reducing costs and improving the efficiency of healthcare delivery. In conclusion, a heart disease prediction research project has significant potential to improve public health by identifying individuals at high risk of developing heart disease and personalizing preventive interventions. With the availability of large amounts of health data and advances in machine learning and artificial intelligence techniques, now is an opportune time to pursue this research. The objective of this research project is to devise a highly accurate method for predicting heart disease.

1.3 Contribution

The key contribution of this paper are as follows:

- 1) Developed a more reliable and comprehensive data set, by merging four datasets.
- 2) Six classification models - Logistic Regression, Decision Tree, Random Forest, Extreme Gradient Boost, Naive Bayes, and K-Nearest Neighbors (KNN) - were employed to predict heart disease.
- 3) Additionally, the stacking ensemble technique was utilized to enhance the testing rate and minimize execution time.
- 4) Finally, the overall results were used to assess and compare the performance of the various models.

1.4 Organization of the paper

The remaining sections of the paper are structured as follows. In Section 2, we'll go over the suggested reading lists and contributions of a variety of authors. In Section 3, we detail the proposed methodology. The datasets that this paper draws upon are described in Section 4. In Section 5, we describe the findings and analyze the relevant current works. In Section 6, we provide an overview of the study and draw a conclusion from our research.

2 Related works

In this article, we have mostly discussed the efforts of numerous researchers who have developed methods for identifying heart attacks using machine learning. Combining RF with LM has been shown to yield better outcomes in previous research. Therefore, the HRFLM approach was proposed, which ended up being 88.47% accurate [2]. Earlier research suggested combining chi-square (CHI) with principal component analysis (PCA) to improve the accuracy of machine learning models used to forecast cardiovascular disease. Accuracy of 98.7% was attained on the Cleveland dataset, 99.0% on the Hungarian dataset, and 99.4% on the CH dataset using the proposed technique [4]. Methods for disease prediction were evaluated based on how well they performed using many different supervised machine learning algorithms. The investigations found that RF had the most accuracy 53% of the time, whereas SVM had the highest accuracy 41% of the time [5]. When the dataset size is small, R. Bharti et al. [1] showed that machine learning should be used. Machine learning approaches performed better than statistical methods, according to Ramesh TR et al. [7]. Experiments were undertaken by F.S. Alotaibi [8] to improve classifier performance by the use of numerous machine learning models to maximize data usage. Research has shown that the Relief feature selection process can produce a highly correlated feature set that works well with various ML methods. The most important aspects of RFBM were likewise found to perform remarkably well in the study [9]. Using the Cleveland Heart Disease dataset, J.P. Li et al. [10] created an effective machine learning-based diagnostic system for heart disease by selecting features with Fast Conditional Mutual Information (FCMIM) and validating the system with a Leave-One-Subject-Out (LOSO) validation technique. Expert system for heart failure diagnosis using stacked SVMs was proposed

by L. Ali et al. [11]. The system makes use of two SVM models, one with a linear kernel and the other with a radial basis function (RBF) kernel. The purpose of this research was to find a way to improve the diagnostic process for heart failure. Using a majority voting approach, C.B.C. Latha et al. [12] analyzed the accuracy of heart disease prediction using an ensemble of classifiers consisting of NB, BN, RF, and MP. Mukesh Kumar et al. [13] presented idea to remove collinearity and underperforming characteristics from the chosen data set before using it for ensemble modelling. Saihood et al. [14] proposed a model that combines several machine learning algorithms to improve their prediction power by using ensemble approaches including bagging, boosting, and stacking. M.S. Reza et al. [15] suggested two stacking-based models for the categorization of diabetes utilizing a combination of deep neural network stacking ensemble techniques and conventional methods to merge the predictions of various classification models. In Table 1, work related to heart attack prediction using several machine learning techniques are being presented.

3 Proposed methodology

Preprocessing the data in the study's initial phase entails dealing with missing values, scaling, and encoding categorical variables. The data is then divided into training and testing sets for the creation and assessment of the model. The study employs six classification methods: Naive Bayes, Extreme Gradient Boost, Decision Tree, Logistic Regression, and K-Nearest Neighbours (KNN). Various criteria, including accuracy, precision, recall, and F1-score, are used to assess each classification technique's performance. In the second portion of the study, ensemble techniques—specifically stacking—are used to boost the effectiveness of the classification models. A machine learning ensemble technique called stacking combines numerous models to increase the precision of a prediction. A meta-classifier that has been trained on the base model predictions is used in stacking to integrate the predictions of many base models. The outcomes demonstrate that stacking considerably enhances the performance of the base classifiers. The suggested work's flowchart is shown in Fig. 1. The algorithm of proposed method is presented below in Table 8.

3.1 Preprocessing

Data preparation is essential before data analytics. The model-training dataset may be noisy or inconsistent. Data may potentially be missing. To address anomalies, data preparation is necessary. A well-known basic linear regression imputation method is used to handle missing values [16]. The technique employs the correlation matrix to establish attribute value correlation and then predicts missing values using existing attribute values in a dataset. Normalization converts numeric attributes in the dataset to a similar scale without distorting ranges or losing information. This minimizes machine learning algorithm complexity and processing cost.

3.2 Feature selection

In the feature selection [17] stage, the goal is to pull the features out of the collection. The benefits of the Feature Extraction stage are that it improves accuracy, lowers the risk of over-fitting, speeds up training, makes it easier to see the data, and makes the model easier

Table 1 Work related to heart attack prediction using several machine learning techniques

Author	Dataset	Techniques used	Contribution	Result
Bharti et al. [1]	Public Health Dataset which consists of four databases: Cleveland, Hungary, Switzerland, and Long Beach V	Lasso, LR XGBoost, sequential model with fully connected dense layer.	validated the use of ML even when the dataset is small.	KNN's performance can be improved by normalizing the data and incorporating outlier detection and statistical analysis
Mohan et al. [2]	Cleveland dataset from a UCI machine learning repository	NB, GLM, LR, DL, DT, RF, GBT and SVM	Integrated RF into LM and suggest the HRFMLM approach for enhanced performance.	Accuracy = 88.47%
Garate-Escamila et al. [4]	Cleveland, Hungarian, and a combination of both called CH (Cleveland-Hungarian)	CHI-PCA with RF	advanced combining the chi-square (CHI) and principal component analysis (PCA) to boost ML prediction accuracy.	The accuracy rates are as follows: 98.7% for Cleveland, 99.0% for Hungarian, and 99.4% for CH.
Uddin et al. [5]	Scopus and PubMed dataset	Supervised ML algorithms: SVM, LR, ANN, DT, RF, NB, KNN.	Evaluation of Supervised Machine Learning Algorithms for Disease Prediction	RF had highest accuracy in 53% of studies, SVM in 41%.
Ramesh et al. [7]	UCI Center Machine Learning database	Supervised classification techniques: SVM, KNN, DT, RF, NB.	ML outperformed statistical techniques.	RF has 88.35% prediction rate
Alotaibi et al. [8]	UCI Heart Disease Data Set	DT, NB, LR, RF, SVM.	Multiple ML models used to give better classification	Decision Tree- 93.19%
Ghosh et al. [9]	Five datasets (Cleveland, Hungary, Switzerland, VA Long Beach, and Statlog) were merged.	Lasso and relief to extract features.	Relief feature selection gives a correlated feature set for ML, RFBM works well for high-impact features.	Accuracy of RF using Relief feature selection algorithm = 97.89%
Li et al. [10]	Cleveland Heart Disease	FCMIM feature selection and LOSO validation.	ML diagnosis system created	FCMIM-SVM = 92.37% accuracy
Ali et al. [11]	Cleveland heart disease	Two SVM models created with different regularization and kernels (linear and RBF).	To aid in the diagnosis of heart failure, a stacked support vector machine (SVM) based expert system was presented.	L1 linear SVM + L2 linear and RBF SVM- 92.22% accuracy

Table 1 (continued)

Author	Dataset	Techniques used	Contribution	Result
Latha et al. [12]	Cleveland heart dataset from the UCI	Majority vote with NB, BN, RF and MP	Examines the performance of a collection of classifiers in predicting cardiac issues.	85.48% accuracy
Mukesh Kumar et al. [13]	PIMA	Fish swarm algorithm, three layered artificial neural networks	The idea presented is to remove collinearity and underperforming characteristics from the chosen data set before using it for ensemble modelling.	88.3% accuracy
Saihood et al. [14]	PIMA	Stacked Random Forest with SVM, XGB	Proposed a model that combines several machine learning algorithms to improve their prediction power by using ensemble approaches including bagging, boosting, and stacking.	97.50% accuracy with stacked random forest with SVM
Reza et al. [15]	PIMA	Deep NN both classical and stacking ensemble method, 5-fold CV	Suggested two stacking-based models for the categorization of diabetes utilizing a combination of deep neural network stacking ensemble techniques and conventional methods to merge the predictions of various classification models.	95.50% accuracy

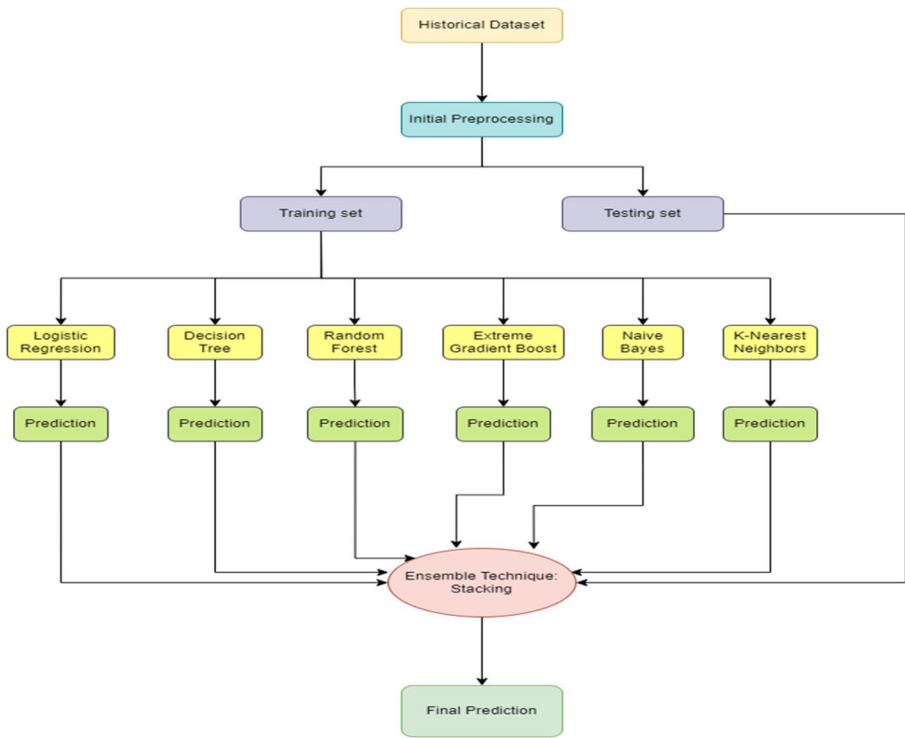


Fig. 1 Workflow of proposed stacking based ensemble model

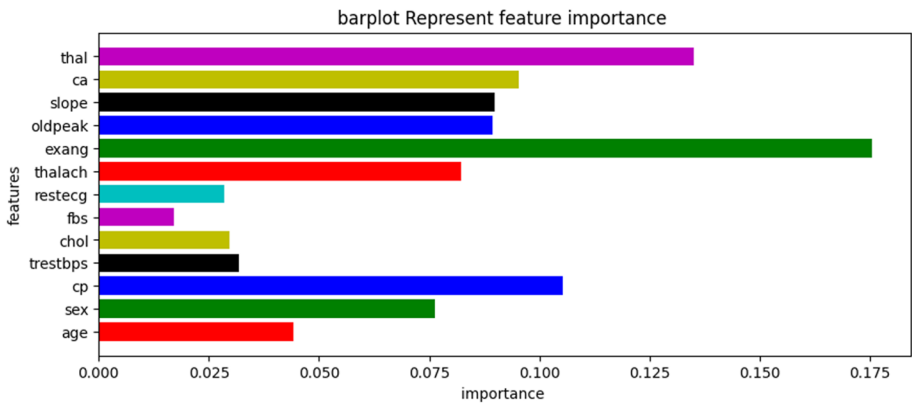


Fig. 2 Presenting the importance of features for heart disease

to explain. Figure 2 shows the importance of features in determining the stage of heart disease of any patient. Out of 14 features, some are very important as shown in the graph. Exercise Induced Angina is the most important feature in determining heart condition followed by maximum heart rate achieved.

3.3 Classification

A key issue in machine learning is classification [6, 18], which entails giving a label or class to an input based on its features or qualities. The output is typically a discrete label that denotes the category or class to which the input belongs. The input is typically a series of numerical or categorical data that describe an object or event. In supervised learning, when the machine learning algorithm is trained on a labelled dataset made up of input-output pairs, classification is frequently utilized. The program uses a range of methods, including decision trees, support vector machines, logistic regression, and neural networks, to learn to map the input properties to the matching output labels. Binary classification and multiclass classification are the two primary categories of classification issues. The output variable in a binary classification can have one of two possible values, such as true or false or positive or negative. Problems involving binary classification include those involving fraud detection, medical diagnosis, and spam detection. The output variable in multiclass classification can be one of more than two alternative values, such as various species of animals, illnesses, or goods. Image identification, text classification, and sentiment analysis are a few examples of multiclass classification issues. Typically, parameters like accuracy, precision, recall, F1 score, and area under the ROC curve are used to evaluate classification models. The evaluation metric you use depends on the particular issue at hand as well as the cost of committing various errors. For instance, false negatives (failing to diagnose an illness) may be more expensive than false positives (diagnosing a disease when it is not existent) in a medical diagnosis task. Classification is a potent machine learning technology with numerous uses, including image recognition, audio recognition, natural language processing, recommendation systems, and many others. It is a crucial method for resolving issues in numerous fields and companies.

3.3.1 Logistic regression

Logistic regression [18] is a commonly used statistical technique in machine learning, specifically for binary classification tasks. This regression analysis method predicts the likelihood of a binary outcome based on a set of input predictors or variables. By using maximum likelihood estimation, it calculates the coefficients of the predictors and transforms them into probabilities through the logistic function. The predicted probabilities fall between 0 and 1, indicating the probability of the outcome variable being in one of the two classes. Equation (1) shows the equation for logistic regression where $p(x)$ represents the probability of an output given the input x , β_0 represents the intercept or bias, β_1 represents the weight or coefficient associated with the input feature. Its working algorithm is written in Table 2.

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}} \quad (1)$$

3.3.2 Decision tree

For both classification and regression, the decision tree [19] is a popular machine learning approach. It's a model that looks like a tree that uses the values of the input features to divide the data into smaller and smaller chunks. Each feature or attribute is represented by

Table 2 Algorithm for logistic regression

Algorithm: Logistic Regression Input: Training set represented as Train_set with feature x and class label y Output: Logistic Regression Model. Procedure: Step – 1: for i from 1 to N and $x \in X$ and $y \in Y$ Apply equation (1) on the training dataset Step-2: Using maximum likelihood estimation, calculate coefficients β_0 and β_1 of the predictors. Step-3: transform calculated coefficients into probabilities through logistic function, $p(x)$ Step – 4: Return probability values Step -5: Make prediction based on obtained probability values as 1 for heart attack 0 for not heart attack end
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a node inside the model, and each projected value or class is represented by a node at the model’s leaves. A criterion such as entropy or Gini impurity is used to create the split, with the goal of creating homogeneous subsets. The technique works well with numerical and categorical information and is easy to understand. In Eq. (2), p_{ij} can represent either the probability of position or the probability of the negative class, and this is the entropy equation. Its working algorithm is written in Table 3.

$$Entropy = - \sum_{j=1}^m p_{ij} \log_2 p_{ij}$$

(2)

3.3.3 Random forest

The random forest [17, 20] is an effective machine learning algorithm that uses a group of decision trees to improve prediction accuracy. It constructs several decision trees on random subsets of the data, then combines them to make a final prediction. By training each tree on a random subset of input features, the algorithm reduces overfitting and improves

Table 3 Algorithm for decision tree

Algorithm: Decision Tree Input: Training set represented as Train_set with feature x and class label y Output: Decision Tree Model Procedure: Step – 1: for i from 1 to N and $x \in X$ and $y \in Y$ start forming tree for entire dataset Step -2: Calculate entropy for each node using equation (2) Step-3: Based on entropy value select the best unused attribute Step – 4: Split given dataset in subset of data using selected attribute Step - 5 Repeat the process for each subset till no further classification takes place Step- 6: Find the leaf node of the tree, this will give prediction end
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generalization. Moreover, the algorithm employs bootstrapping, a technique that generates numerous data samples with replacement to increase the diversity of the trees. The random forest is a versatile algorithm that can handle regression and classification problems, as well as noisy and high-dimensional data. Its working algorithm is written in Table 4.

3.3.4 Extreme Gradient Boost

Extreme Gradient Boosting (XGBoost) [17] is a widely used machine learning algorithm that is employed in regression and classification tasks. It is an optimized version of the gradient boosting algorithm that utilizes a tree-based methodology. XGBoost is created to enhance the performance of gradient boosting by decreasing computation time and enhancing accuracy. It accomplishes this by exploiting parallel processing, regularization, and handling missing values in the data. The algorithm operates by progressively adding decision trees to the model while minimizing a loss function. Its working algorithm is written in Table 5.

3.3.5 Naive Bayes

The Naive Bayes algorithm [18, 21] is a probabilistic classification technique based on Bayes' theorem, which assumes the independence of features. The model calculates the probability of a data point belonging to a certain class given its features. To perform classification, the algorithm needs labeled training data to estimate the conditional probabilities of each feature for each class. The probabilities are then used to classify new data by determining the probability of each class and selecting the one with the highest probability. Naive Bayes is computationally efficient, works well with high-dimensional data, and is easy to implement. Equation (3) describes the posterior probability of class A given predictor B where $P(A)$ is the prior probability of class, $P(B|A)$ is the likelihood which is the

Table 4 Algorithm for random forest

Algorithm: Random Forest Classifier
Input: Training set represented as *Train_set* with feature *x* and class label *y*
Output: Random Forest Classifier Model.
Procedure:
 for *i* from 1 to *N* and $x \in X$ and $y \in Y$ Construct trees considering set of features to form forest.
x: Sample of Test
T: Total trees
n: Number of features considered
end for
 Implementing Final Classifier Decision
for *I* from 1 to *T* :
 Train_set $\leftarrow$ bootstrapofSamples(*Train_set*)
 B $\leftarrow$ buildingTreeClassifier(*Train_set*, *n*)
end for
 Calculate the number of votes by applying to each tree
 Return class with highest vote
end

Table 5 Algorithm for Extreme Gradient Boost

Algorithm: XGBoost
Input: θ , instance set of node
Input: $\theta_k = \{i \in \theta | x_{ik} \neq \text{missing}\}$
Input: d , feature dimension
 Also applies to the approximate setting, only collect statistics of non – missing entries into buckets

$gain \leftarrow 0$
 $G \leftarrow \sum_{i \in I} g_i, H \leftarrow \sum_{i \in I} h_i$

for $k = 1$ **to** m **do**
 //enumerate missing value goto right
 $G_L \leftarrow 0, H_L \leftarrow 0$
 for j in sorted(I_k , ascent order by x_{jk}) **do**
 $G_L \leftarrow G_L + g_i, H_L \leftarrow H_L + h_i$
 $G_R \leftarrow G - G_L, H_R \leftarrow H - H_L$
 $score \leftarrow \max(score, \frac{G_L^2}{H_L + \beta} + \frac{G_R^2}{H_R + \beta} - \frac{G^2}{H + \beta})$
 end
 //enumerate missing value goto Left
 $G_R \leftarrow 0, H_R \leftarrow 0$
 for j in sorted(I_k , ascent order by x_{jk}) **do**
 $G_R \leftarrow G_R + g_i, H_R \leftarrow H_R + h_i$
 $G_L \leftarrow G - G_R, H_L \leftarrow H - H_R$
 $score \leftarrow \max(score, \frac{G_L^2}{H_L + \beta} + \frac{G_R^2}{H_R + \beta} - \frac{G^2}{H + \beta})$
 end
end

probability of the predictor given class and $P(B)$ is the prior probability of the predictor. It working algorithm is written in Table 6.

$$P\left(\frac{A_i}{B_i}\right) = \frac{P\left(\frac{B_i}{A_i}\right) * P(A_i)}{P(B_i)} \quad (3)$$

3.3.6 K-Nearest Neighbors

K-Nearest Neighbors (KNN) [18, 22] is a machine learning algorithm that is widely used for regression and classification problems. It is a non-parametric method that predicts the class of a new data point by finding the K most similar data points in the training set and choosing the most frequent class among them. The choice of distance metric to measure the similarity between data points may vary, with Euclidean and Manhattan distance being popular choices. Although KNN is easy to understand and can handle both categorical and

Table 6 Algorithm for Naive Bayes

Algorithm: Naive Bayes Input: Training set represented as Train_set with feature x and class label y Output: Naive Bayes Model Procedure: Step – 1: for i from 1 to N and $x \in X$ and $y \in Y$ Convert the given dataset features into frequency table Step-2: Using frequency table create likelihood by getting probability of given features Step-3: Calculate the posterior probability using equation (3) Step – 4: Return probability values Step -5: Make prediction based on obtained probability values as 1 for heart attack 0 for not heart attack end
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numerical data, it may be sensitive to the value of K and may become computationally expensive for large datasets. Its working algorithm is written in Table 7.

3.4 Ensemble technique

To boost forecast accuracy and lessen overfitting, ensemble techniques [23, 24] merge numerous models into a single one. These techniques produce multiple models and then combine their results into a single forecast. The acronym bagging describes a specific form of ensemble method known as bootstrap aggregation. Multiple models are trained using various bootstrap samples from the full dataset. Weights, biases, and hyper parameters for each model are independently trained on a part of the data. The aggregate forecast is determined by tallying all the votes cast among the various models. Boosting is another type of ensemble method; it involves training a series of models that each learn to fix the flaws in

Table 7 Algorithm for K-Nearest Neighbors

Algorithm: K-Nearest Neighbors Input: Training set represented as Train_set with feature x and class label y Output: K – Nearest Neighbors Model. Procedure: Step – 1: for i from 1 to N and $x \in X$ and $y \in Y$ Select K numbers of neighbors Step-2: Calculate the Euclidean distance of K neighbors Step-3: From calculated Euclidean distance select K nearest neighbors Step – 4: Number of points in every category need to be counted among these K neighbors Step - 5 Newly data points must be assigned to that category for which neighbors are maximum Step- 6: Final model is ready end

the prior model. Decision trees are frequently used as the basis for boosting models; many models are trained on the same data but are given varying weights based on the faults of the prior models. Predictions from multiple models are averaged to reach a single estimate, with greater emphasis placed on more precise models. One further method for combining numerous models in a hierarchical fashion is stacking, an ensemble method. This strategy involves training a number of models on the training data and then using their predictions as features in a more advanced model. This more sophisticated model is trained to integrate the outputs of more basic models. When the strengths and shortcomings of the underlying models are compatible, stacking can be an extremely effective technique. Machine learning models can benefit greatly from the use of ensemble approaches, which can increase their accuracy even when the individual models themselves have large variation or are prone to overfitting. However, they can make the model more difficult to train and more computationally intensive to deploy. Which ensemble method is best for a given problem is conditional on the nature of the data and the models being employed.

3.4.1 Stacking

By training a meta-model that learns how to integrate its predictions, stacking is an ensemble strategy that combines many machine learning models. This meta-model learns how to translate the predictions of the underlying models to the final output by using those predictions as input features. In a stacking technique, the data is often divided into several folds, with each fold being used to train a set of base models. Consider dividing the data into four folds and using the four base models A, B, C, and D. Model A would be trained on folds 2–4, Model B on folds 1, 3, 4, Model C on folds 1, 2, 4, and Model D on folds 1–3. We use the base models to generate predictions on a validation set that was not used for training after training them on their respective folds. The input features for the meta-model are then taken from the predictions made by the basic models on the validation set. Any model, including a logistic regression, decision tree, or neural network, can be the meta-model. To learn how to combine the predictions of the basis models to generate a final prediction, the meta-model is trained on the predictions of the base models as well as the true labels of the validation set. Once trained, the meta-model can be used to predict outcomes based on fresh, unexplored data. The meta-model predictions are utilized as input characteristics for a higher-level meta-model in practice, allowing the stacking technique to be used repeatedly. This can result in further accuracy gains, but it also makes the model more complex and demands more computer power. Especially when the base models have complementary strengths and weaknesses, stacking is a potent ensemble strategy that can increase the accuracy of machine learning models. To prevent overfitting and to make sure that the meta-model is learning to integrate the base models in a meaningful way, however, careful tweaking and validation are necessary (Table 8).

4 Dataset

The study uses the Public Health Dataset (<https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset?datasetId=216167&sortBy=voteCount>), which has 76 characteristics and is made up of four databases: Cleveland, Hungary, Switzerland, and Long Beach V. But in written experiments, only 14 attributes are used, including the predicted attribute that shows whether a patient has heart disease or not, with 0 meaning no disease and 1

Table 8 Algorithm for Proposed Stacking based Ensemble technique

Algorithm: Stacking based proposed ensemble method

Input: Historical dataset containing heart disease controlling attributes

Output: Possibility of heart attack or not

Procedure:

Step – 1 for all input dataset, split into training and testing dataset

Step – 2 Preprocess both training and testing data

Step – 3 Apply all six classifiers such as LR, DT, RT, XGBoost, NB and KNN to makes initial prediction over Training dataset with extracted features.

Step – 4 Stack all these classifier to build meta classifier.

Step – 5 Evaluate model performance

End

meaning disease. Age, sex, type of chest pain, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results, maximum heart rate, exercise-induced angina, ST depression, slope of the peak exercise ST segment, number of major vessels, thalassemia, and the target attribute are all mentioned. Each attribute has a specific use or meaning. For example, the patient's age is given in years, and the type of chest pain has four different values. Description of dataset's attribute is given in Table 9.

5 Result and discussion

The results of our suggested model's individual steps are discussed below. Sensitivity, precision, specificity, accuracy, F1 score, etc. described in Table 10 are just a few of the performance indicators that can be used to assess a model's efficacy. The model, dataset,

Table 9 Dataset attribute details

S.No	Attribute name	Description	Range of values
1	age	age in years	29 to 77
2	sex	Gender of the person [1: Male, 0: Female]	0, 1
3	cp.	Chest pain type [0-Typical Type 1 Angina; 1- Atypical Type Angina; 2-Non-angina pain; 3-Asymptomatic]	0, 1, 2, 3
4	trestbps	Resting Blood Pressure in mmHg	94 to 200
5	chol	Serum cholesterol in mg/dl	126 to 564
6	fbs	fasting blood sugar > 120 mg/dl (1 = true; 0 = false)	0, 1
7	restecg	resting electrocardiographic results	0, 1, 2
8	thalach	maximum heart rate achieved	71 to 202
9	exang	Exercise Induced Angina	0, 1
10	oldpeak	ST depression induced by exercise relative to rest	0 to 6.2
11	slope	the slope of the peak exercise ST segment	0, 1, 2
12	ca.	number of major vessels (0–3) colored by fluoroscopy	0 to 3
13	thal	1 = normal; 2 = fixed defect; 3 = reversible defect	1 to 3
14	target	Class attribute (1 = disease; 0 = No risk of heart disease)	0, 1

challenge, and context all play a role in deciding which performance indicators to utilize. Below, we go over the indicators we utilized to judge the effectiveness of our suggested methodology. Table 11 presents the result of different classifiers. Figures 3 and 4 shows confusion matrices and Receiver Operating Curves of different classifiers respectively. The Confusion matrix generated in our proposed model is presented in Fig. 5.

Table 12 shows values obtained by the proposed technique for heart disease predictions. In Table 13, we have compared the accuracy obtained by all six individual classifiers and proposed stacking based ensemble technique. We can see that, in terms of accuracy, stacking based ensemble technique outperforms. Figure 6 shows RoC for all the individual models along with proposed model. It is seen clearly that the area under curve value of proposed model is highest, which suggests our model is performing better than the existing ones.

The study covers 1025 participants with 14 features, including age, sex, chest pain kind, blood pressure, cholesterol, and other clinical parameters. The study begins by addressing missing values, scaling, and encoding categorical variables. For model creation and evaluation, data is separated into training and testing sets. The study uses Logistic Regression, Decision Tree, Random Forest, Extreme Gradient Boost, Naive Bayes, and K-Nearest Neighbors (KNN). Accuracy, precision, recall, and F1-score are used to evaluate classification techniques. Random forest and decision tree are most accurate at 92.68%, followed by Extreme Gradient Boosting at 90.73%. The second portion of the study uses stacking to increase classification model performance. Ensemble machine learning 5-fold stacks models to increase prediction accuracy. A trained meta-classifier combines the predictions of numerous basic models in stacking. Stacking considerably enhances base classifier performance. The stacked model outperforms all classifiers with 98.53% accuracy. In Table 14, we have compared our proposed work with different existing works. From the table, it is very much clear that our proposed model is giving more accurate result in terms of accuracy than existing machine learning based models. Figure 7 represents accuracy obtained by different classifiers graphically. Figure 8 Graph showing achieved percentage classification accuracy in various articles along with proposed method.

We compared our findings to previous research on heart disease datasets conducted by various authors using various approaches and presented graphically in Fig. 4. For feature selection, R. Bharti et al. [1] employed the Lasso algorithm. The classifiers used in machine learning are LR, KNN, DT, RF, SVM, and XGBoost. Deep learning was accomplished using a sequential model with a fully connected dense layer. They demonstrated that when the dataset is small, we should apply machine learning. S. Mohan et al. [2] coupled the RF method with the LM and created the HRFLM method, which yielded an accuracy of 88.47%. Garate-Escamila A.K. et al. [4] proposed combining chi-square (CHI) with PCA to improve machine learning model prediction. It achieved 98.7% accuracy for the Cleveland dataset, 99% accuracy for the Hungarian dataset, and 99.4% accuracy when the Cleveland and Hungarian datasets were combined. S. Uddin et al. [5] employed SVM, LR, ANN, DT, RF, NB, and KNN supervised machine learning methods. In 53% of the studies, RF had the highest accuracy. In 41% of the studies, SVM had the highest accuracy. Ramesh TR et al. [7] employed supervised machine learning algorithms such as SVM, KNN, DT, RF, and NB. RF had the highest accuracy of 88.35%. Alotaibi, F.S [8]. employed DT, RF, LR, SVM, and NB and determined that DT provided the maximum accuracy (93.19%). P. Ghosh et al. [9] extracted features with Lasso and Relief and classified them with RF. It has the potential to achieve a maximum accuracy of 97.89%. Fast conditional mutual information (FCMIM) feature selection and the LOSO validation technique were employed by J.P. Li et al. [10]. With FCMIM and SVM, they attained the greatest accuracy of 92.37%.

Table 10 Description of different performance evaluation metrics used

Performance metric	Description	Formula
Accuracy	It shows the proportion of accurate forecasts to all the model's predictions	$Accuracy = \frac{TP+TN}{(TP+FN)+(FP+TN)}$
Sensitivity/Recall (TPR)	The proportion of true positives to all positives is known as sensitivity	$Sensitivity = \frac{TP}{TP+FN}$
Precision	To evaluate accuracy, compare the true positive rate with the projected positive rate	$Precision = \frac{TP}{(TP+FP)}$
Specificity (TNR)	Possible characterizations include the TN/total negatives ratio.	$Specificity = \frac{TN}{TN+FP}$
$F_1 - score$	It evaluates the accuracy of two different classifiers.	$f_1 - score = \frac{2 * TP}{2 * TP + FP + FN}$
Matthews Correlation Coefficient	A high score can only be achieved if the forecast was accurate across all four dimensions of the confusion matrix, making it a more credible statistical rate.	$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP+FP) * (TP+FN) * (TN+FP) * (TN+FN)}}$
AuC	The area under the receiver operating characteristic curve (AUC) indicates how well a model distinguishes between positive and negative examples.	
RoC	When evaluating binary classification jobs, the receiver operating characteristic (ROC) curve is frequently used as a statistic. At various threshold values, it is a curve that compares the genuine positive rate and the false positive rate.	

Table 11 Result obtained in different classifier

Model	Accuracy	F1-Sore	Recall	Precision	Sensitivity	Specificity	MCC	AuC
LR	0.8439	0.8620	0.9090	0.8196	0.8439	0.8620	0.8196	0.9204
NB	0.8439	0.8608	0.9	0.8250	0.8439	0.8608	0.825	0.9185
KNN	0.8585	0.8687	0.8727	0.8648	0.8585	0.8687	0.8648	0.9304
DT	0.9268	0.9289	0.8909	0.9702	0.9268	0.9289	0.9702	0.9702
RF	0.9268	0.9333	0.9545	0.9130	0.9268	0.9333	0.9130	0.9734
XGB	0.9073	0.9132	0.9090	0.9174	0.9073	0.9132	0.9174	0.983
Proposed	0.9853	0.9861	0.9727	1.0	0.9853	0.9861	1.0	0.988

L.Ali et al. [11] employ two SVM, Linear L1 regularised and L2 regularised, with different kernels such as linear and radial basis function (RBF) kernels. They had the best accuracy of 92.22%. C.B.C.Latha et al. [12] achieved an accuracy of 85.48% using NB, SVM, BN, MP and majority voting. Our proposed technique is 5-fold stacking ensemble machine learning which increases classification model performance. A trained meta-classifier combines the predictions of numerous basic models in stacking. Stacking considerably enhances base classifier performance. The stacked model outperforms all classifiers with 98.53% accuracy.

6 Conclusion

This study uses numerous categorization approaches and stacking as ensemble strategies to forecast cardiac disease. The study illustrates the efficacy of combining classifiers to increase heart disease prediction accuracy. The study's findings may help healthcare providers identify patients at risk of heart disease and design focused strategies to manage the disease. The goal of this study is to offer a unique method for predicting heart disease in patients by combining six classification techniques: logistic regression, decision trees, random forests, extreme gradient boosting, naive bayes, and K-nearest neighbors (KNN). The suggested method also employs ensemble techniques, notably stacking, to improve the classifiers' accuracy. The researchers employ several metrics, such as confusion matrix, accuracy, precision, specificity, sensitivity, and F1 score on the dataset, to evaluate the performance of the classifiers. The study discovered that stacking enhances classification accuracy considerably. The researchers intend to apply their proposed strategy to a larger dataset in the future and to investigate various feature selection techniques to improve accuracy even further. Furthermore, the findings and methodology of the study can be applied to other biological disorders, contributing to the larger research topic of employing machine learning in healthcare. Our proposed 5-fold stacking-based ensemble learning technique outperforms all the existing models with 98.53% accuracy. Overall, this study suggests a viable method for predicting cardiac disease using machine learning techniques, with the potential to improve medical decision-making and patient outcomes.

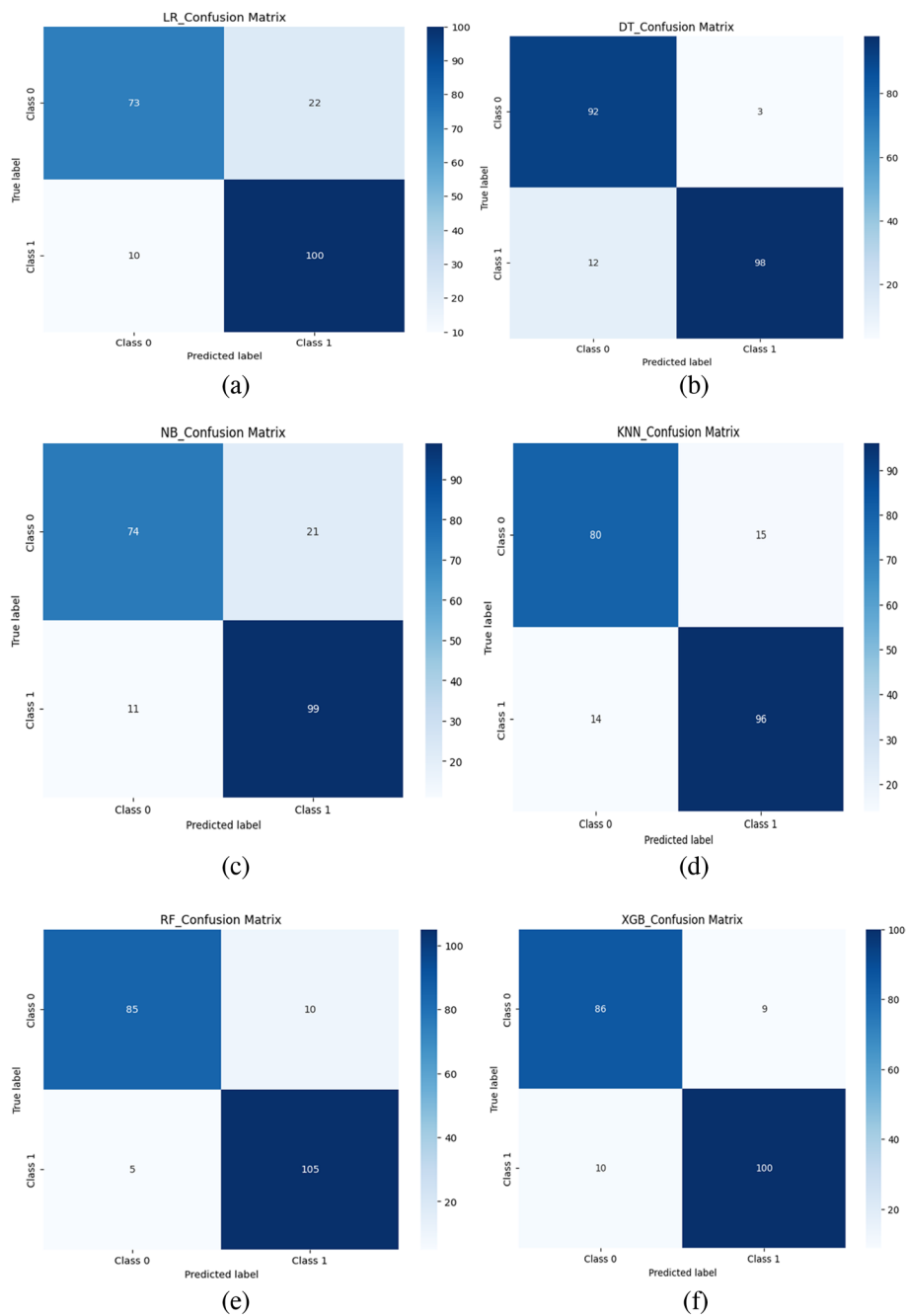


Fig. 3 Confusion matrix for classifier (a) Logistic Regression, (b) Decision Tree, (c) Naïve Bayes, (d) KNN, (e) Random Forest and (f) Extreme Gradient Boost

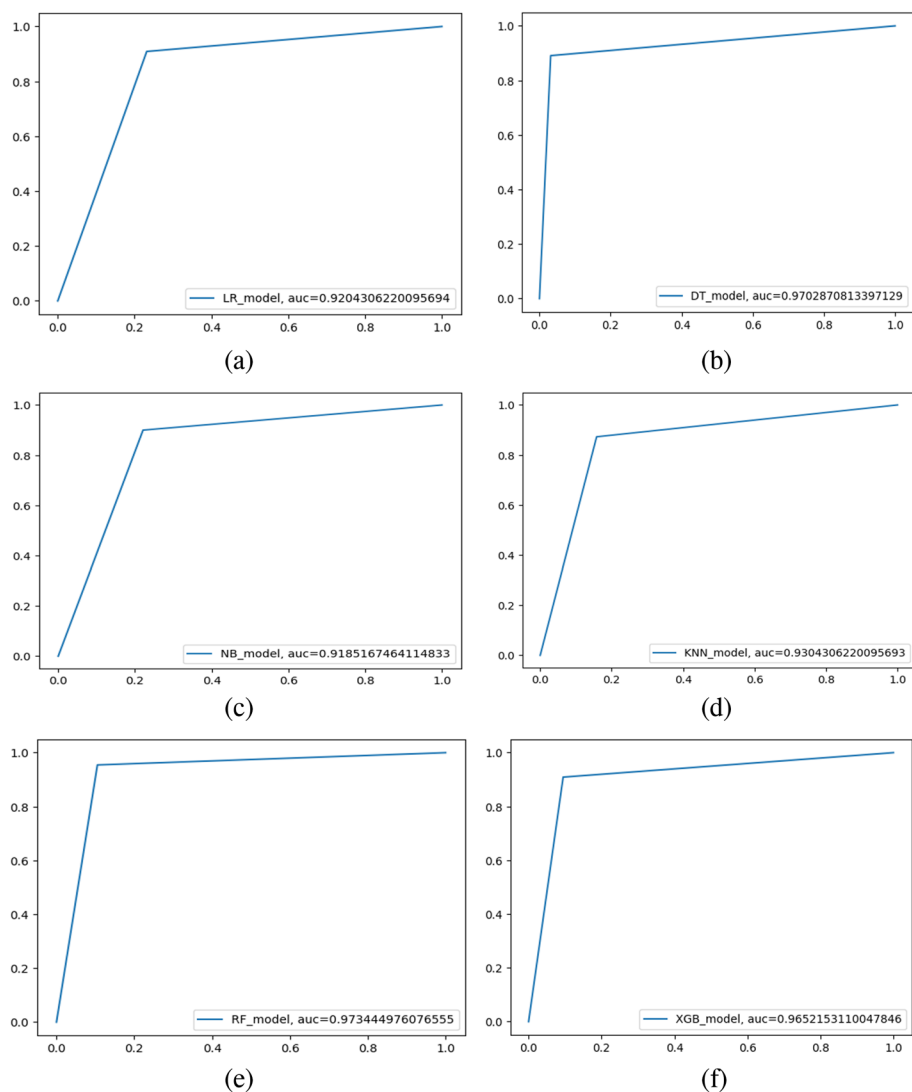


Fig. 4 Receiver Operating curve for classifier (a) Logistic Regression, (b) Decision Tree, (c) Naïve Bayes, (d) KNN, (e) Random Forest and (f) Extreme Gradient Boost

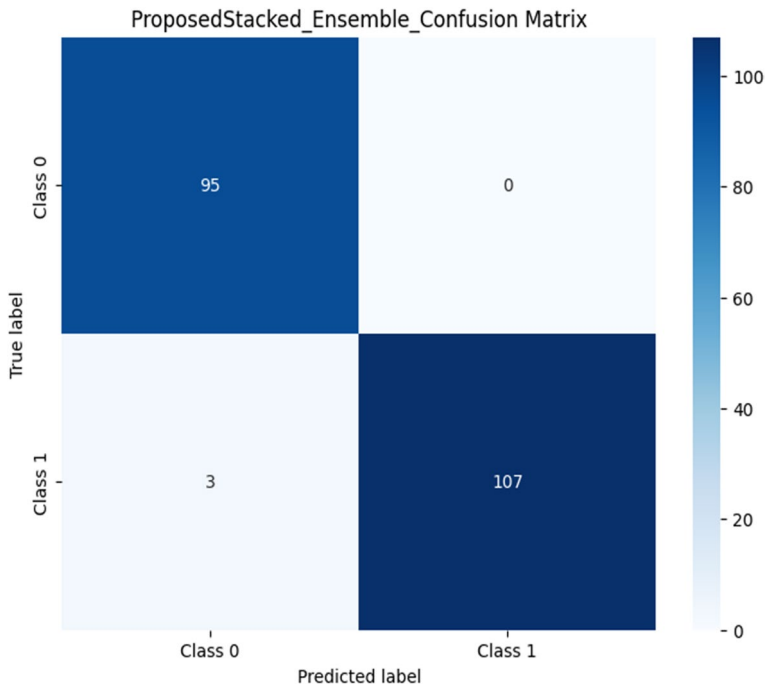


Fig. 5 Confusion matrix for proposed Stacking based ensemble model

Table 12 Performance evaluation of proposed model

Performance metric	Value
Sensitivity	0.9853
Specificity	0.9861
Precision	1.0000
Accuracy	0.9853
F1 Score	0.9861
AuC	0.9880
Matthews correlation coefficient	1.0000

Table 13 Accuracy comparison of individual models with proposed model

S. No	Model	Accuracy
1	Logistic Regression	84.39%
2	Decision Tree	92.68%
3	Random Forest	92.68%
4	Extreme Gradient Boost	90.73%
5	Naive Bayes	84.39%
6	K-Nearest Neighbors	85.85%
7	Stacking based ensemble technique	98.53%

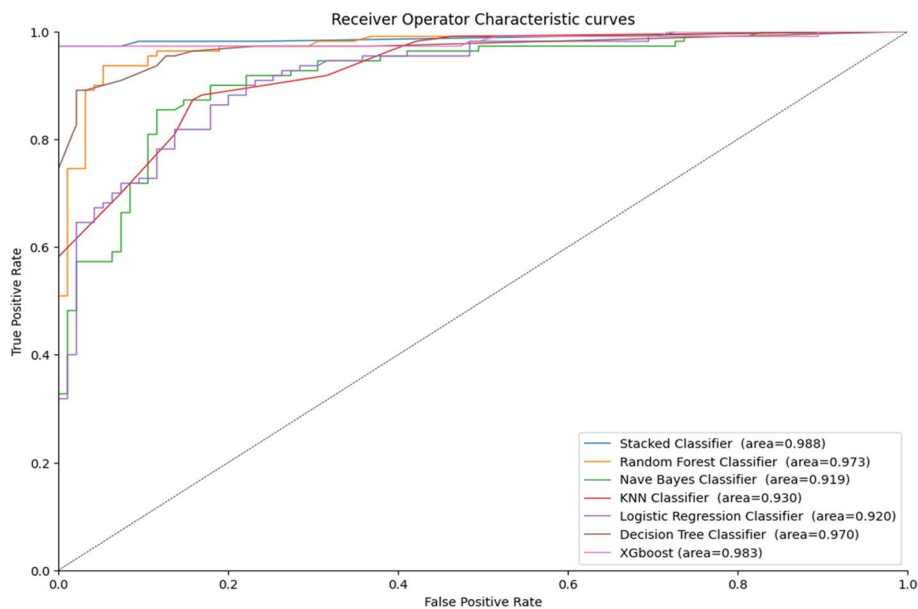


Fig. 6 RoC curve for proposed model

Table 14 Performance comparison of various techniques present in different articles and proposed method

Author	Techniques	Percentage classification accuracy
Mohan et al. [1]	NB, GLM, LR, DL, DT, RF, GBT and SVM	88.47%
Ramesh et al. [5]	SVM, KNN, Decision Tree Classifier, RF, and Naïve Bayes.	88.35%
Alotaibi [6]	NB, LR, DT, RF, and SVM	93.19%
Li et al. [8]	FCMIM-SVM	92.37%
Ali et al. [9]	L1 linear SVM + L2 linear and RBF SVM	92.22%
Latha et al. [10]	Majority vote with NB, BN, RF and MP	85.48%
Proposed technique	5-fold Stacking based ensemble technique	98.53%

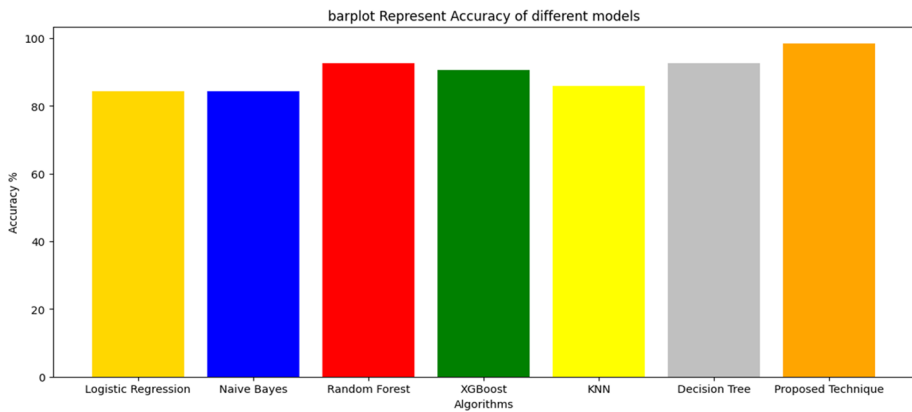


Fig. 7 Graph showing accuracy comparison of individual models with proposed model

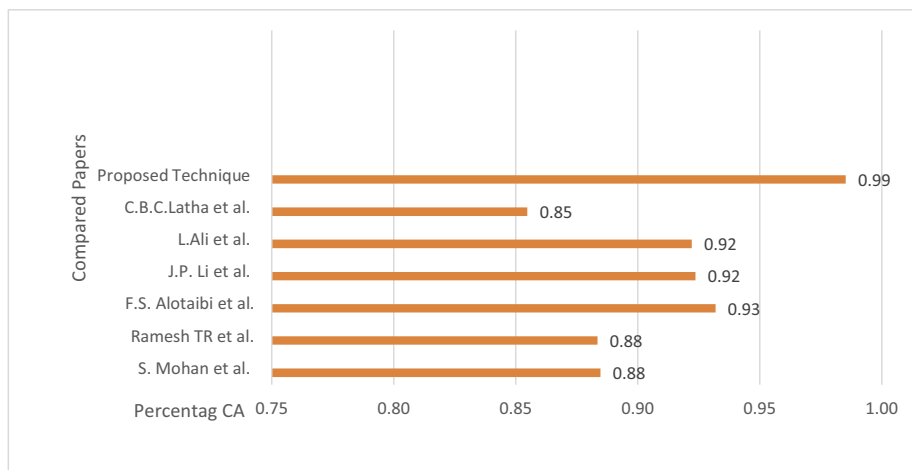


Fig. 8 Graph showing achieved percentage classification accuracy in various articles along with proposed method

Abbreviations IoT: Internet of Things; XGBoost: Extreme Gradient Boosting; ML: Machine Learning; KNN: K-Nearest Neighbors; RF: Random Forest; LM: Linear Model; PCA: Principal Component Analysis; CHI: Chi-square; CH: Cleveland-Hungarian; SVM: Support Vector Machine; RFBM: Random Forest Bagging Method; FCMIM: Fast Conditional Mutual Information; LOSO: Leave-One-Subject-Out; HRFLM: Hybrid Random Forest with Linear Model; LASSO: Least Absolute Shrinkage and Selection Operator; NB: Naive Bayes; BN: Bayesian Network; MP: Multilayer Perceptron; GLM: Generalized Linear Model; LR: Logistic Regression; DL: Deep Learning; DT: Decision Tree; GBT: Gradient Boosted Trees; ANN: Artificial neural network; NB: Naive Bayes; TP: True Positive; TN: True Negative; FP: False Positive; FN: False Negative; RBF: Radial Basis Function

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Declarations

Conflict of interest There are no conflicts of interest declared by the authors.

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